



KG-1 BOOK A

Abacus Mental Arithmetic

Student Information

Name:

Level:

Date:

3

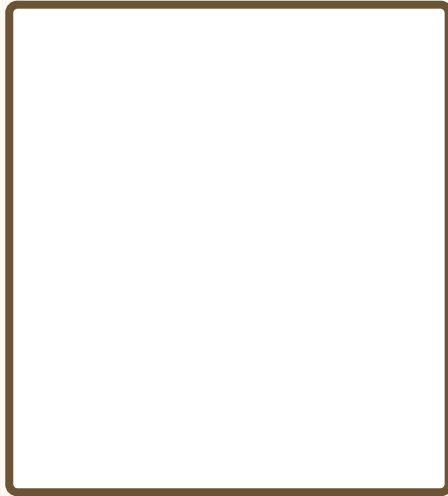
5

4

Description of the Abacus

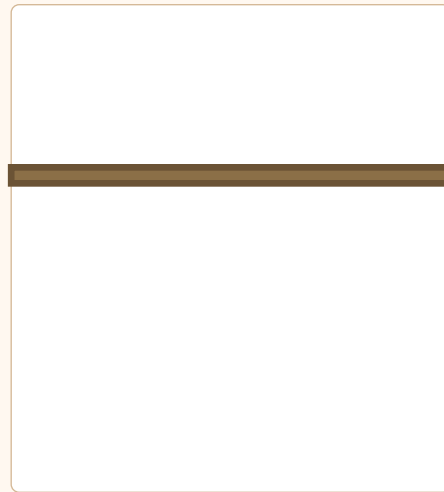
Frame

The outer rectangle



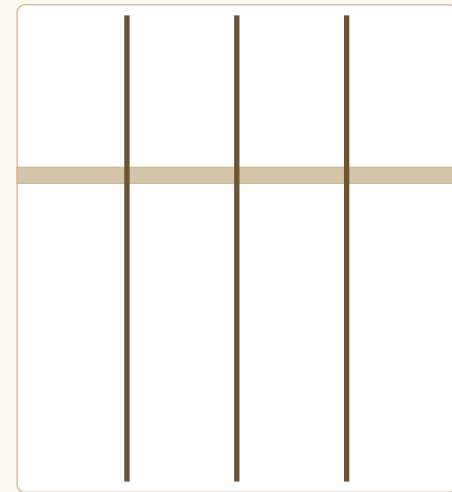
Beam

Horizontal divider bar



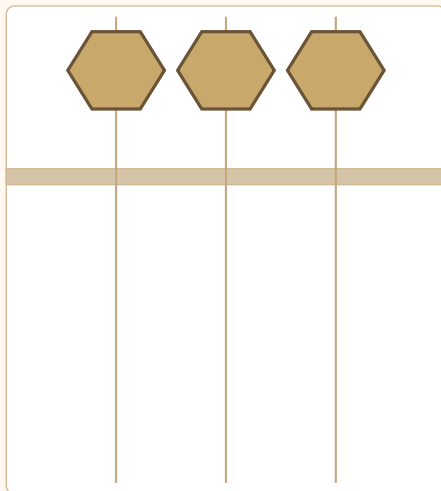
Rod

Vertical bars for beads



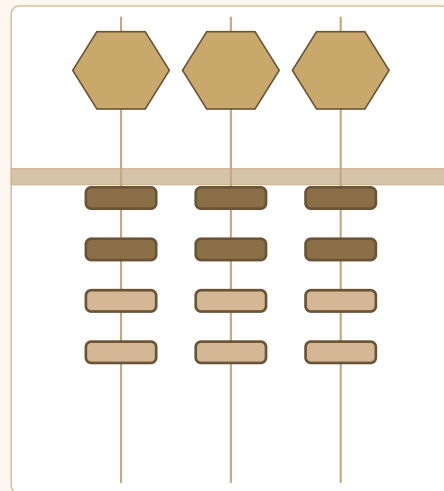
Upper Bead

Value = 5



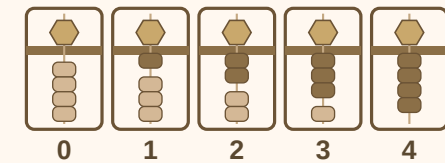
Lower Bead

Value = 1 each



Complete Abacus

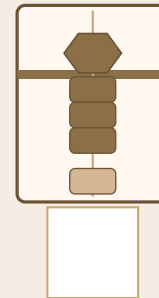
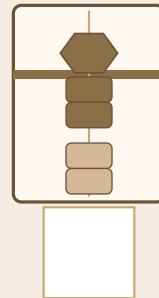
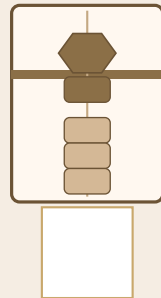
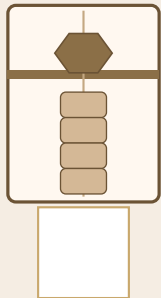
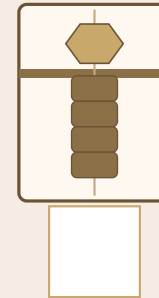
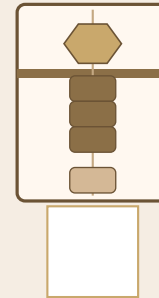
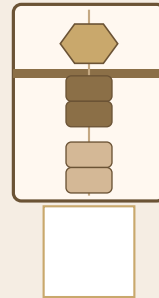
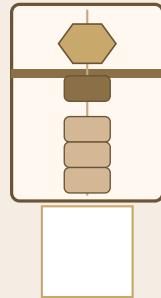
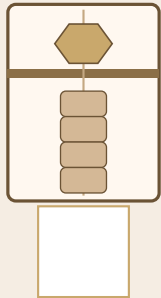
All parts together: values 0 to 9



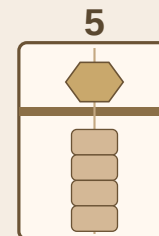
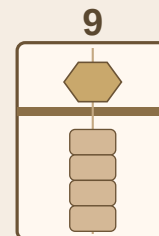
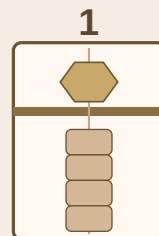
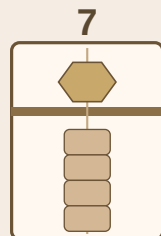
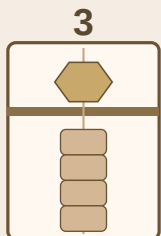
Write the Value of Beads

Set 1 to 9 on the abacus

Look at each abacus and write the number it shows in the box below.



Now set these numbers on the abacus:



Understand Numbers

Numbers 1 to 5



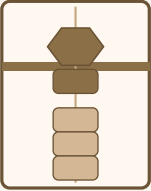
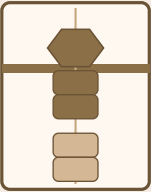
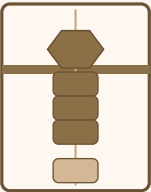
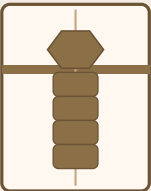
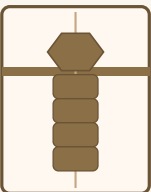
Abacus	Hand	Dots	Number	Trace	Trace	Practice
			1	1	1	
			2	2	2	
			3	3	3	
			4	4	4	
			5	5	5	



Understand Numbers

Numbers 6 to 10



Abacus	Hand	Dots	Number	Trace	Trace	Practice
	 5+1		6	6	6	
	 5+2		7	7	7	
	 5+3		8	8	8	
	 5+4		9	9	9	
	 5+5		10	10	10	



Write the Numbers You Can Identify

How many numbers can you find hidden in the picture? Write them in the boxes!



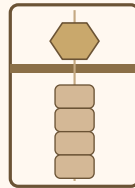
Write the numbers you found:

Count and Write

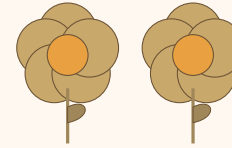
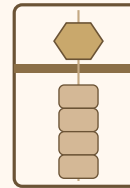
Count the objects, write the number, then show it on the abacus.



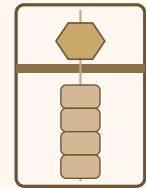
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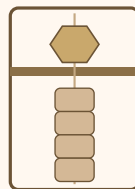
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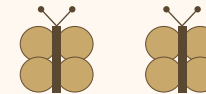
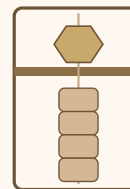
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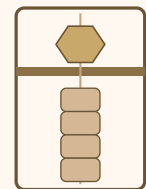
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=

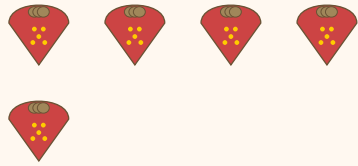


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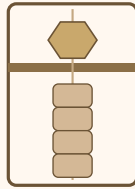


Count and Write

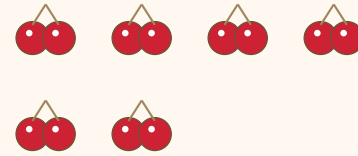
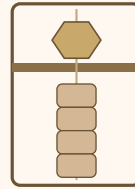
Count the objects, write the number, then show it on the abacus.



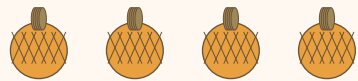
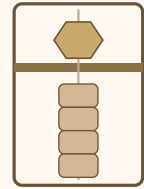
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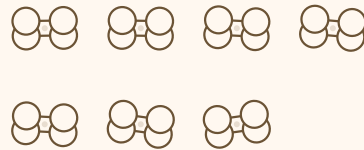
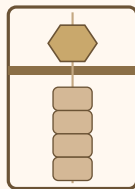
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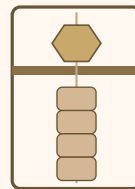
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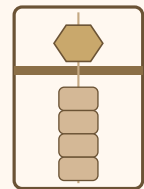
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=



=



Match the Following

Draw lines to match each number with its dot pattern and abacus.

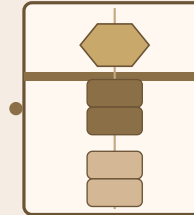
Number

Dots

Abacus

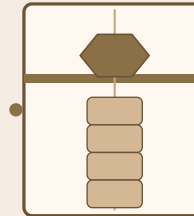
1

••••



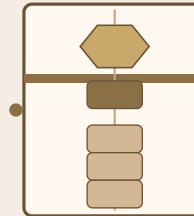
2

•••••



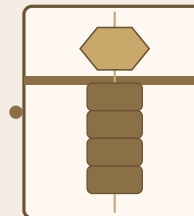
3

•••



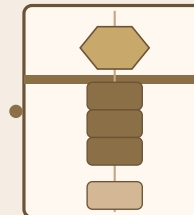
4

•



5

••



Match the Following

Draw lines to match each number with its dot pattern and abacus.

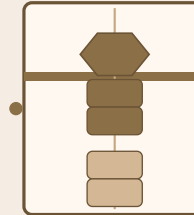


Number

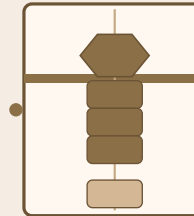
Dots

Abacus

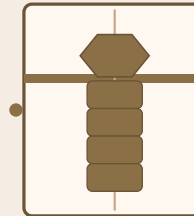
6



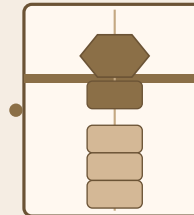
7



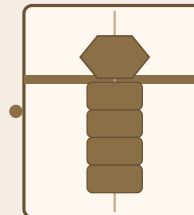
8



9



10



Abacus Calculation: 1 Digit 3 Rows



Fingering Ex: 1-1, 2-1, 3-1, 4-1, 2-2, 3-2

S.No.	1	2	3	4	5	6	7	8
1	1	2	3	4	1	3	4	2
2	1	-1	-1	-2	2	-2	-1	1
3	-1	1	-1	1	-1	2	-2	-2
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	1	2	4	3	2	4	1	3
2	3	-1	-3	-1	2	-2	4	1
3	-2	3	1	2	-3	-1	-3	-3
Ans.								

S.No.	1	2	3	4
1	4	2	3	1
2	-1	1	-2	2
3	-2	-3	1	-3
Ans.				

3 + 2 =

5 - 3 =

4 + 1 =

5 - 4 =

Direct Addition & Subtraction



Fingering Ex: 1+2, 2+1, 3+1, 5+1, 2+2, 5+2

S.No.	1	2	3	4	5	6	7	8
1	1	2	3	4	1	2	3	4
2	2	1	1	-2	4	3	2	-3
3	-1	-2	-3	1	-2	-4	-1	2
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	2	4	3	2	4	1	3	4
2	-1	-1	1	2	-2	3	-1	-3
3	3	-2	-2	-3	1	-1	2	1
Ans.								

S.No.	1	2	3	4
1	1	3	2	4
2	2	-2	1	-1
3	1	2	-3	1
Ans.				





2 + 4 =





5 - 2 =





3 + 2 =





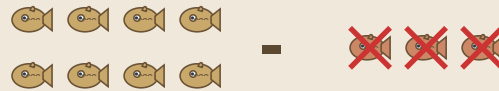
4 - 3 =

Direct Addition & Subtraction

Count the objects and write the answer.



$5 + 3 =$



$8 - 3 =$



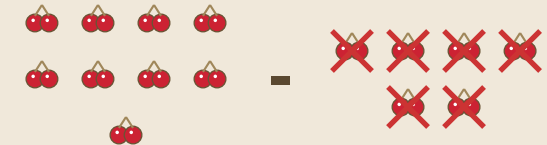
$7 - 5 =$



$4 + 5 =$



$5 + 4 =$



$9 - 6 =$

Abacus Calculation: 1 Digit 3 Rows



Fingering Ex: 1+5, 2+5, 3+5, 4+5, 5-5, 6-5

S.No.	1	2	3	4	5	6	7	8
1	1	2	3	4	1	2	3	4
2	6	5	5	5	7	6	5	5
3	-2	-3	-4	-6	-5	-1	-1	-2
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	7	4	5	6	8	7	5	9
2	-5	-3	-2	-1	-5	-3	3	-5
3	3	6	4	-3	1	-2	-6	-1
Ans.								

S.No.	1	2	3	4
1	1	7	3	8
2	5	-2	5	-5
3	-3	1	-6	2
Ans.				





$1 + 5 =$





$4 - 2 =$





$2 + 3 =$





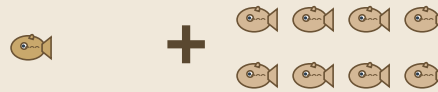
$5 - 1 =$

Direct Addition

Count the objects and write the answer.



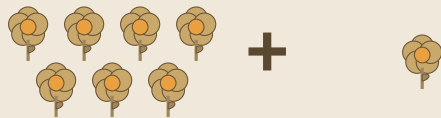
$2 + 6 =$



$1 + 8 =$



$2 + 7 =$



$7 + 1 =$



$3 + 5 =$



$4 + 4 =$



Abacus Calculation: 1 Digit 3 Rows



Fingering Ex: 1+6, 2+6, 3+6, 1+7, 2+7, 1+8

S.No.	1	2	3	4	5	6	7	8
1	1	2	3	4	1	5	6	7
2	6	7	-5	-2	7	-3	-2	-5
3	-5	-2	7	5	-2	1	-3	1
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	1	2	3	4	8	7	6	9
2	5	8	-5	-3	-6	-5	-3	-1
3	3	-6	6	5	3	4	1	-8
Ans.								

S.No.	1	2	3	4
1	1	2	7	6
2	7	8	-5	-5
3	-5	-3	1	1
Ans.				





$3 + 3 =$





$5 - 4 =$





$1 + 4 =$





$4 - 2 =$

Abacus Calculation: 1 Digit 3 Rows



S.No.	1	2	3	4	5	6	7	8
1	3	7	5	2	1	4	6	8
2	4	-2	-3	7	9	-2	-3	-5
3	-5	4	6	-5	-7	5	-2	3
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	4	9	2	5	7	8	3	1
2	6	-4	-1	3	-3	-6	6	8
3	-7	1	8	-6	5	4	-5	-7
Ans.								

*

S.No.	1	2	3	4
1	6	9	4	7
2	-2	-7	5	-1
3	1	2	-3	-5
Ans.				





 $4 + 2 =$





 $5 - 3 =$





 $2 + 3 =$





 $3 - 1 =$

Abacus Calculation: 1 Digit 3 Rows



S.No.	1	2	3	4	5	6	7	8
1	4	8	6	3	7	1	5	2
2	-1	1	-2	-3	-5	8	4	-1
3	-2	-7	5	9	2	-6	-7	6
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	8	6	3	9	4	7	2	5
2	-6	-1	7	-2	5	-3	6	-4
3	7	-3	-4	-5	-8	1	-5	3
Ans.								

S.No.	1	2	3	4
1	9	7	4	8
2	-2	-1	-3	-5
3	1	-5	5	-2
Ans.				



$3 + 1 =$



$4 - 1 =$



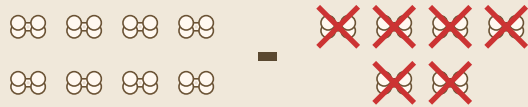
$2 + 4 =$



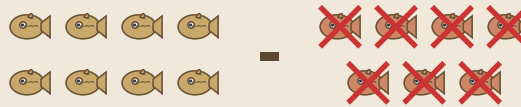
$5 - 2 =$

Direct Subtraction

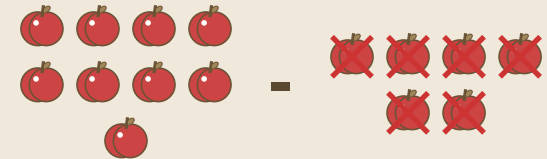
Count the objects and write the answer.



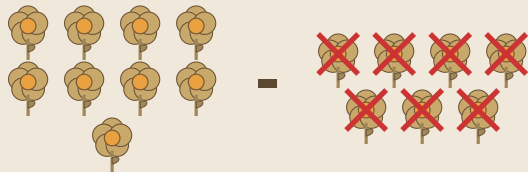
$8 - 6 =$



$8 - 7 =$



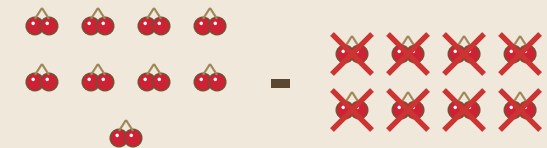
$9 - 6 =$



$9 - 7 =$



$7 - 6 =$



$9 - 8 =$



Abacus Calculation: 1 Digit 3 Rows



Fingering Ex: 6-6, 7-6, 8-6, 9-6

S.No.	1	2	3	4	5	6	7	8
1	7	8	9	6	5	4	3	2
2	-6	-6	-6	-6	3	6	-2	7
3	1	3	-2	7	-6	-8	8	-6
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	6	7	8	9	6	5	4	3
2	1	-3	-6	-6	3	-4	6	7
3	-6	5	4	-1	-7	8	-9	-6
Ans.								

S.No.	1	2	3	4
1	7	9	8	6
2	-6	-6	-6	-6
3	6	3	-1	4
Ans.				

 + 

4 + 1 =

 - 

5 - 2 =

 + 

1 + 3 =

 - 

4 - 3 =

Abacus Calculation: 1 Digit 3 Rows



Fingering Ex: 7-7, 8-7, 9-7, 8-8, 9-8, 9-9

S.No.	1	2	3	4	5	6	7	8
1	7	8	9	8	9	9	7	8
2	-7	-7	-7	-8	-8	-9	-7	-8
3	9	4	-1	5	3	8	6	9
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	9	7	8	9	8	9	7	8
2	-9	-7	-8	-7	-7	-9	-6	-7
3	7	8	6	4	6	5	8	-1
Ans.								

S.No.	1	2	3	4
1	9	7	8	9
2	-8	-7	-8	-9
3	7	9	5	8
Ans.				





$2 + 2 =$





$4 - 3 =$





$3 + 3 =$





$5 - 4 =$

Abacus Calculation: 1 Digit 3 Rows



Fingering Ex: 8-8, 9-8

S.No.	1	2	3	4	5	6	7	8
1	1	7	8	4	9	2	6	3
2	9	-8	-8	-8	-8	8	-8	-8
3	-8	9	5	9	3	-9	7	8
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	9	7	8	5	3	1	4	9
2	-8	2	-7	-8	6	8	-3	-8
3	4	-8	4	9	-8	-7	8	6
Ans.								

S.No.	1	2	3	4
1	8	9	7	8
2	-8	-8	-8	-8
3	7	4	9	1
Ans.				




$5 + 1 =$




$3 - 2 =$




$4 + 2 =$




$5 - 3 =$

Abacus Calculation: 1 Digit 3 Rows



Fingering Ex: 8-8, 9-8, 9-9

S.No.	1	2	3	4	5	6	7	8
1	9	8	7	9	8	7	9	8
2	-9	-8	-7	-8	-7	-6	-9	-8
3	7	9	6	4	3	5	1	2
Ans.								



S.No.	1	2	3	4	5	6	7	8
1	9	8	7	9	8	9	7	8
2	-7	-8	-6	-9	-7	-8	-6	-6
3	4	9	8	7	6	3	5	-1
Ans.								



S.No.	1	2	3	4
1	9	8	7	9
2	-7	-8	-6	-9
3	4	9	8	7
Ans.				



$1 + 4 =$

$5 - 1 =$

$3 + 2 =$

$4 - 2 =$



Composition of 5

Understanding how numbers make 5

5 can be split into two parts in different ways!

5

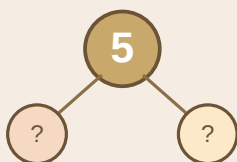
is made of two parts

4 & 1  = 5

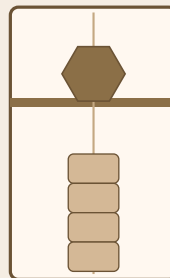
3 & 2  = 5

2 & 3  = 5

1 & 4  = 5

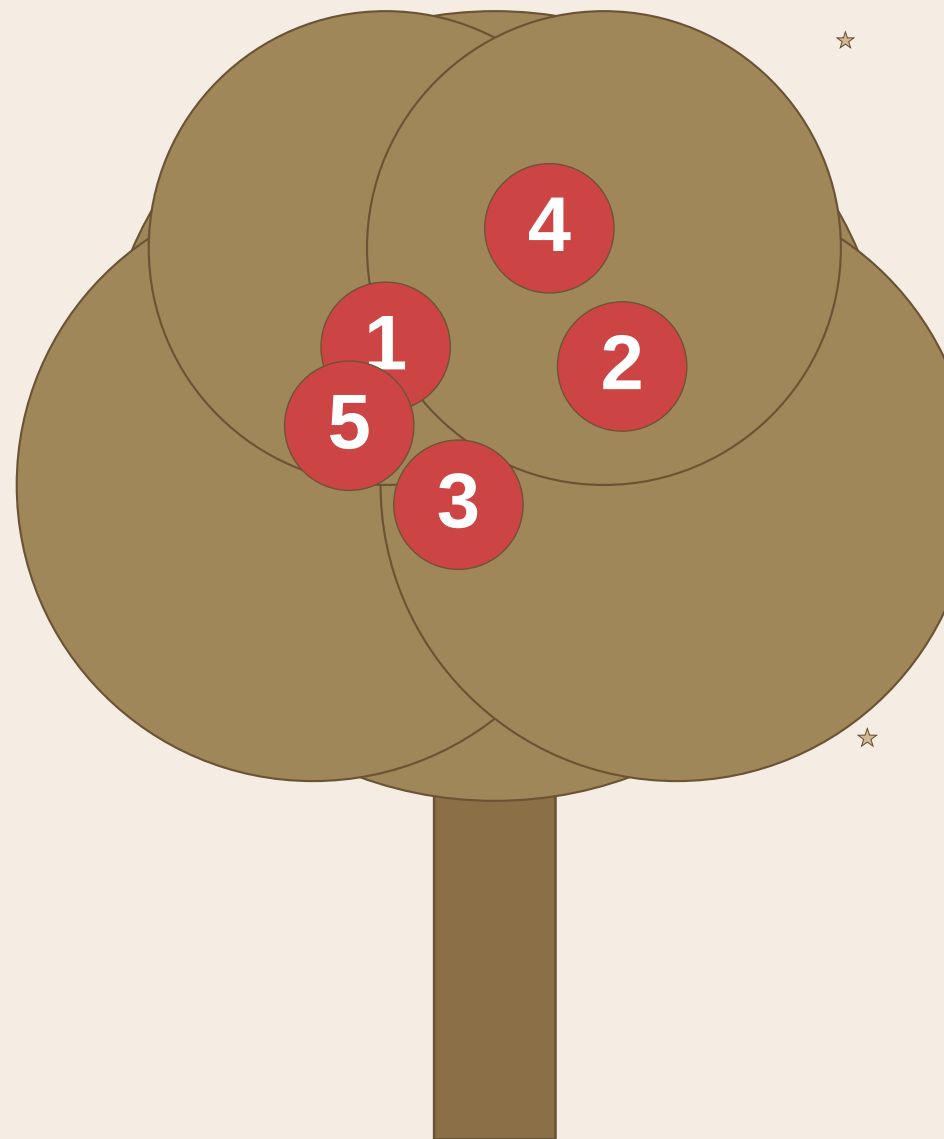


Number bond: split 5 into two parts



5

On the abacus,
5 is one upper bead.



Five apples on the tree!



Fill in the missing number:

4 + = 5

 + 3 = 5

1 + = 5

 + 2 = 5

3 + = 5

 + 4 = 5

Composition of 5

Showing combinations on the abacus

See how we make 5 on the abacus!



2 + 3 = 5

2

5

4 + 1 = 5

4

5

1 + 4 = 5

1

5

3 + 2 = 5

3

5



Abacus Calculation: Composition of 5 (+4)



+4=+5-1 | Fingering: 1+4, 2+4, 3+4, 4+4

S.No.	1	2	3	4	5	6	7	8
1	1	2	3	4	1	2	3	4
2	4	4	4	4	4	4	4	4
3	-3	-5	-6	-7	2	-1	-5	-3
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	2	1	3	4	1	2	3	4
2	4	4	4	4	4	4	4	4
3	-3	-5	1	-6	-4	-5	-2	-1
Ans.								

S.No.	1	2	3	4
1	1	4	2	3
2	4	4	4	4
3	-3	-5	-6	1
Ans.				





$2 + 3 =$





$5 - 3 =$





$4 + 1 =$





$3 - 2 =$

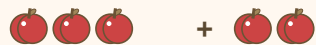
Abacus Calculation: Revision



S.No.	1	2	3	4	5	6	7	8
1	1	2	3	7	4	5	8	1
2	3	6	-1	-5	2	-2	-6	-5
3	-2	-5	4	2	-3	4	3	9
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	2	7	1	4	8	5	9	7
2	4	-5	3	-2	-3	4	-6	2
3	-3	6	5	6	1	-7	2	-8
Ans.								

S.No.	1	2	3	4
1	3	2	6	5
2	4	-1	3	2
3	-5	7	-8	-4
Ans.				



$3 + 2 =$



$4 - 1 =$



$2 + 3 =$



$5 - 2 =$

Abacus Calculation: Composition of 5 (+3)



+3=+5-2 | Fingering: 2+3, 3+3, 4+3

S.No.	1	2	3	4	5	6	7	8
1	2	3	4	2	3	4	2	3
2	3	3	3	3	3	3	3	3
3	-4	-5	-6	1	-1	-5	-3	-4
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	4	2	3	4	2	3	4	2
2	3	3	3	3	3	3	3	3
3	-2	-5	1	-7	-4	-6	-3	-1
Ans.								

S.No.	1	2	3	4
1	3	4	2	3
2	3	3	3	3
3	-5	-4	-3	2
Ans.				





$4 + 2 =$





$5 - 4 =$





$2 + 4 =$





$4 - 1 =$

Abacus Calculation: Revision



S.No.	1	2	3	4	5	6	7	8
1	3	2	4	7	5	8	1	6
2	4	6	-2	-3	3	-5	4	-4
3	-5	-3	5	1	-6	2	-3	3
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	2	4	5	7	1	8	3	9
2	3	-3	-5	-2	6	-3	5	-7
3	-4	6	3	-4	-5	-4	-7	2
Ans.								

S.No.	1	2	3	4
1	4	6	5	7
2	3	-2	-3	-5
3	-5	3	2	4
Ans.				





3 + 3 =





5 - 2 =





1 + 5 =





4 - 3 =

Count and Write



Count the objects on each side, then write the addition and answer.



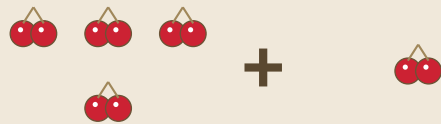
$4 + 2 =$



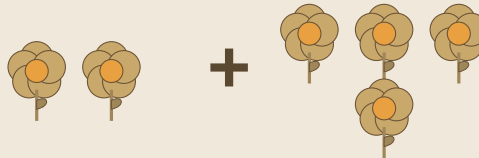
$3 + 2 =$



$2 + 3 =$



$4 + 1 =$



$2 + 4 =$



$3 + 4 =$



Abacus Calculation: Composition of 5 (+2)



+2=+5-3 | Fingering: 3+2, 4+2

S.No.	1	2	3	4	5	6	7	8
1	3	4	3	4	3	4	3	4
2	2	2	2	2	2	2	2	2
3	-4	-5	-3	-6	1	-4	-2	-1
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	3	4	3	4	3	4	3	4
2	2	2	2	2	2	2	-2	2
3	-5	-3	2	-6	-4	-1	4	-7
Ans.								

S.No.	1	2	3	4
1	3	4	3	4
2	2	2	2	2
3	-4	-6	-5	1
Ans.				


 +
 

 $2 + 2 =$


 -
 

 $4 - 2 =$


 +
 

 $3 + 1 =$


 -
 

 $5 - 3 =$

Abacus Calculation: Revision



S.No.	1	2	3	4	5	6	7	8
1	2	4	5	7	3	6	1	8
2	8	3	2	-3	4	-5	7	-4
3	-3	-5	-6	1	-2	4	-3	-2
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	4	3	6	9	2	5	8	1
2	2	5	-4	-5	7	-3	-2	4
3	-3	-7	2	-3	-1	4	-5	3
Ans.								

S.No.	1	2	3	4
1	4	7	5	3
2	3	-5	2	4
3	-2	3	-4	-6
Ans.				





5 + 1 =





3 - 1 =





2 + 3 =





4 - 2 =

Abacus Calculation: Composition of 5 (+1)



+1=+5-4 | Fingering: 4+1

S.No.	1	2	3	4	5	6	7	8
1	4	4	4	4	4	4	4	4
2	1	1	1	1	1	1	1	1
3	-3	-5	2	-4	-2	-6	3	-1
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	4	4	4	4	4	4	4	4
2	1	1	1	1	1	1	1	1
3	-5	6	-3	-5	-2	-4	2	-5
Ans.								

S.No.	1	2	3	4
1	4	4	4	4
2	1	1	1	1
3	-3	-5	-4	2
Ans.				

 +
 

1 + 3 =

 -
 

4 - 1 =

 +
 

3 + 2 =

 -
 

5 - 2 =

Abacus Calculation: Revision



S.No.	1	2	3	4	5	6	7	8	9	10
1	3	7	5	8	2	6	9	4	1	3
2	-4	-2	-5	-6	4	-3	-7	5	7	6
3	6	3	4	3	-5	1	3	-8	-3	-5
Ans.										



S.No.	1	2	3	4	5	6	7	8	9	10
1	5	8	6	3	4	7	2	9	1	5
2	4	-3	-5	7	-2	-6	5	-4	8	-2
3	-7	2	4	-9	5	8	-3	-3	-6	4
Ans.										

S.No.	1	2	3	4	5	6	7	8	9	10
1	4	7	2	8	5	3	6	9	1	4
2	6	-5	4	-3	-4	7	-2	-5	3	5
3	-8	3	-3	1	6	-8	4	-2	-4	-7
Ans.										







 $4 + 2 =$





 $5 - 3 =$





 $2 + 2 =$





 $3 - 1 =$

Abacus Calculation: Revision

All Aboard the Speedy Scholars Express!



Solve each problem and write answers in the wheel circles!

SPEEDY SCHOLARS

Problem	Answer
$3 \times 4 = \underline{\quad}$	Ans
$1 \times 3 = \underline{\quad}$	Ans
$2 \times 2 = \underline{\quad}$	Ans
$4 \times 1 = \underline{\quad}$	Ans